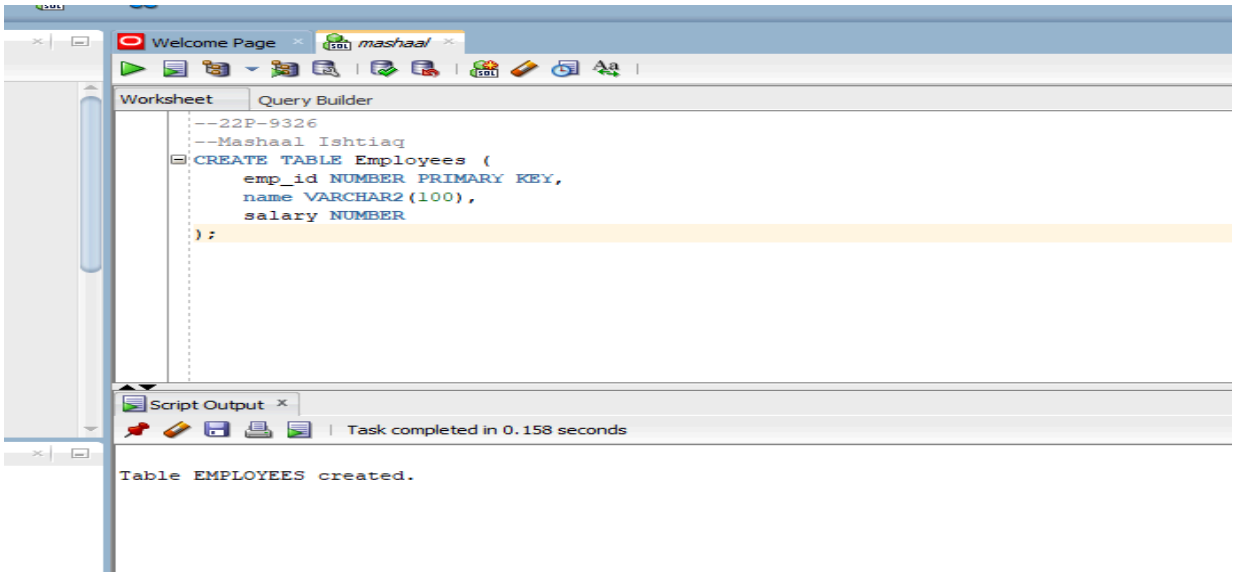
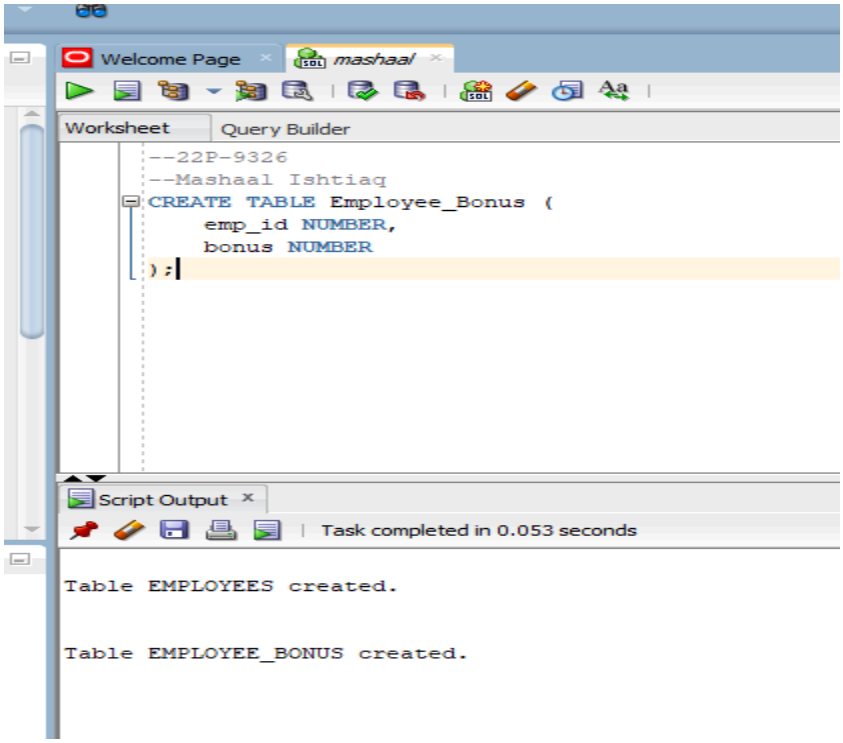


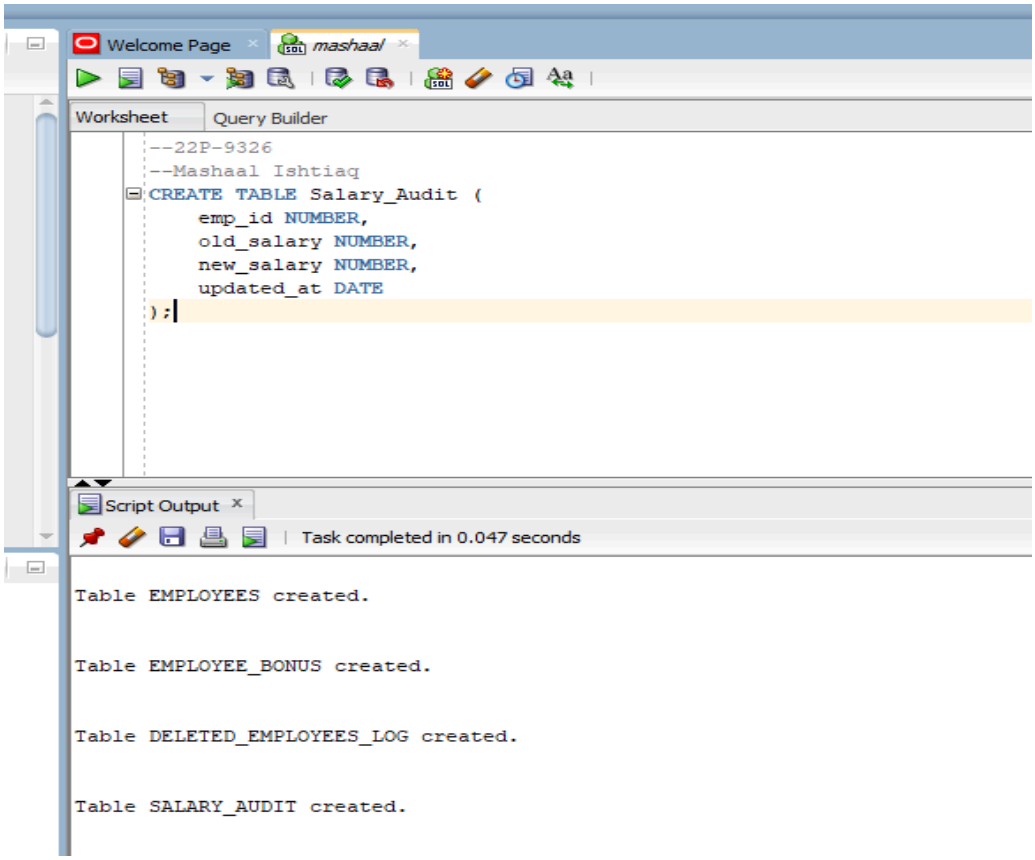
DATABASE LAB

lab#9

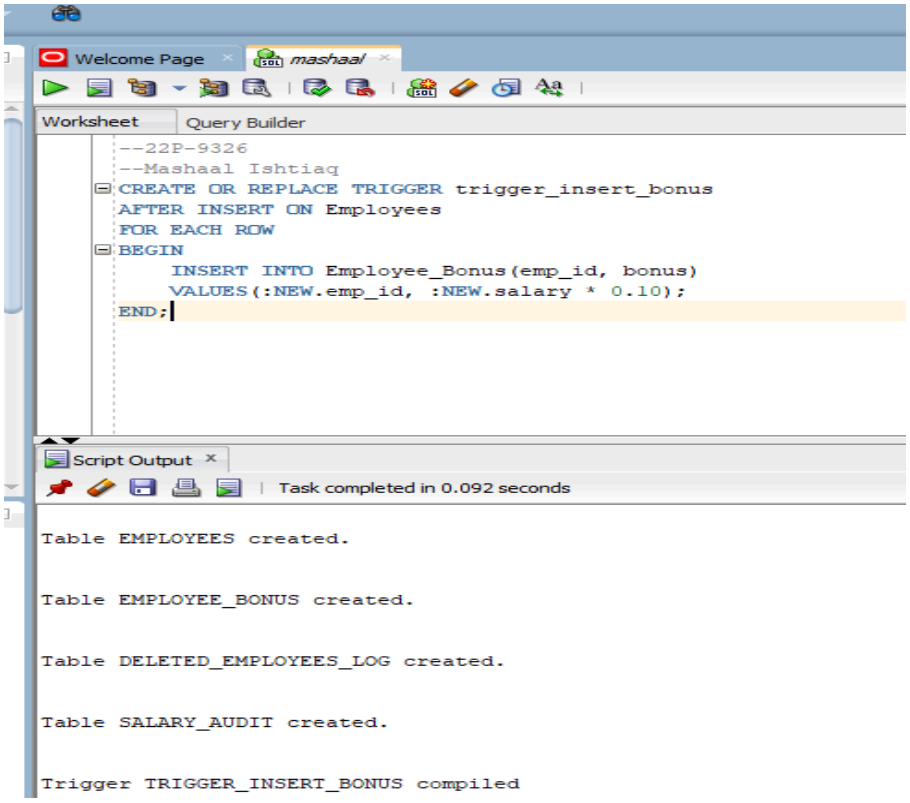
NAME : MASHAAL ISHTIAQ
ROLL NO: 22P-9326
SECTION: BCS-6B


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Employees (
emp_id NUMBER PRIMARY KEY,
name VARCHAR2(100),
salary NUMBER
);
Script Output
Task completed in 0.158 seconds
Table EMPLOYEES created.

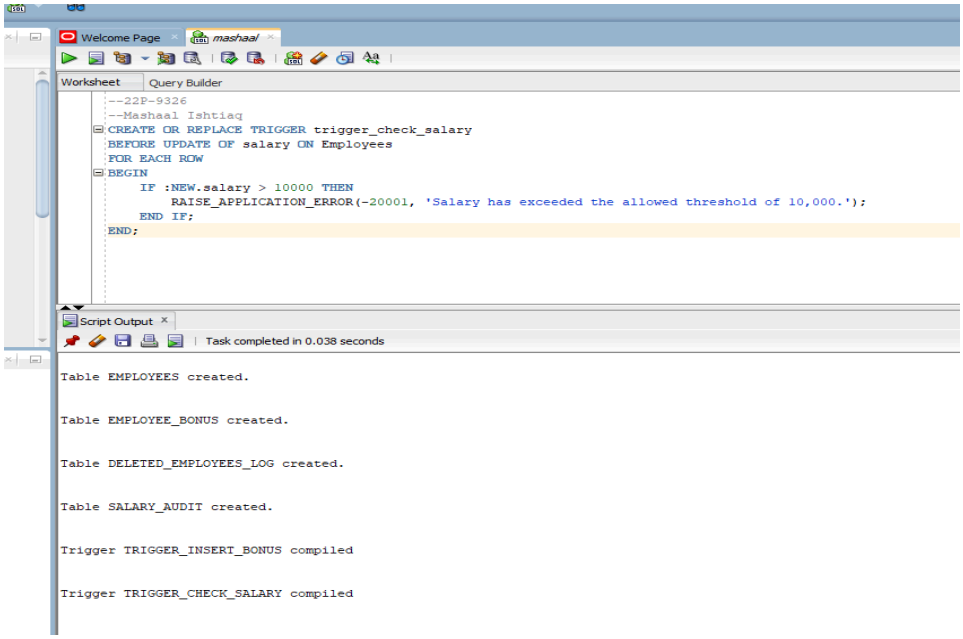

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Employee_Bonus (
emp_id NUMBER,
bonus NUMBER
);
Script Output
Task completed in 0.053 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Salary_Audit (
emp_id NUMBER,
old_salary NUMBER,
new_salary NUMBER,
updated_at DATE
);
Script Output
Task completed in 0.047 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.
Table DELETED_EMPLOYEES_LOG created.
Table SALARY_AUDIT created.

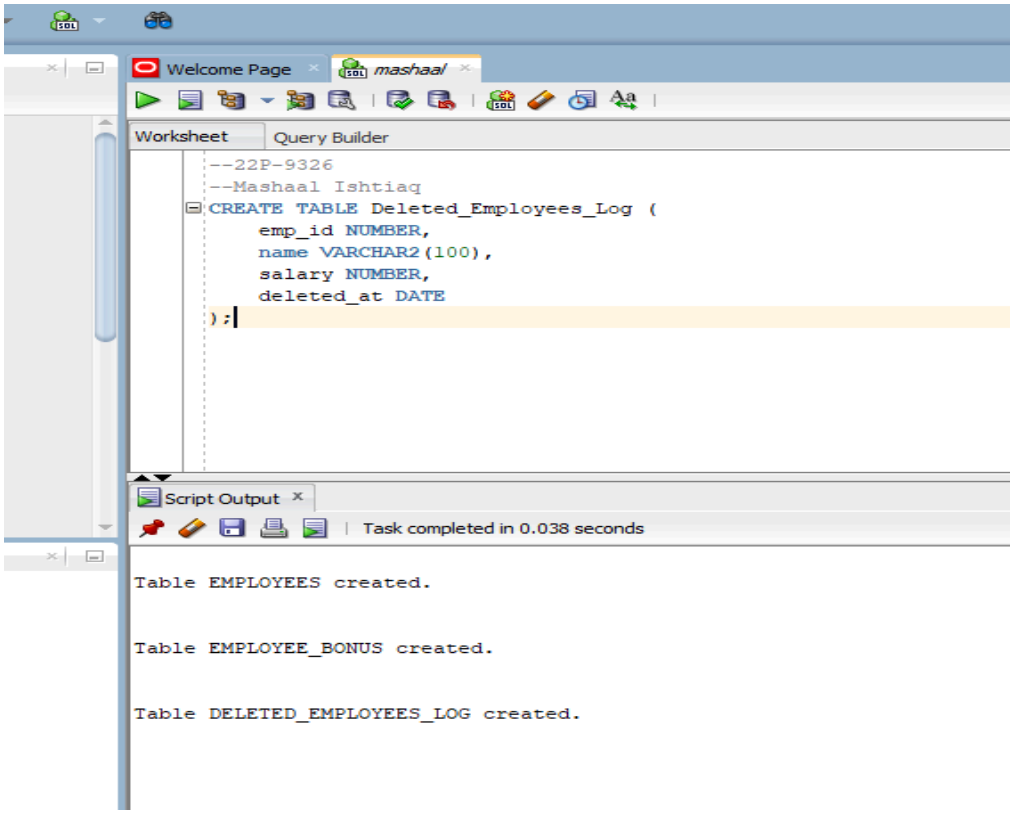
#1

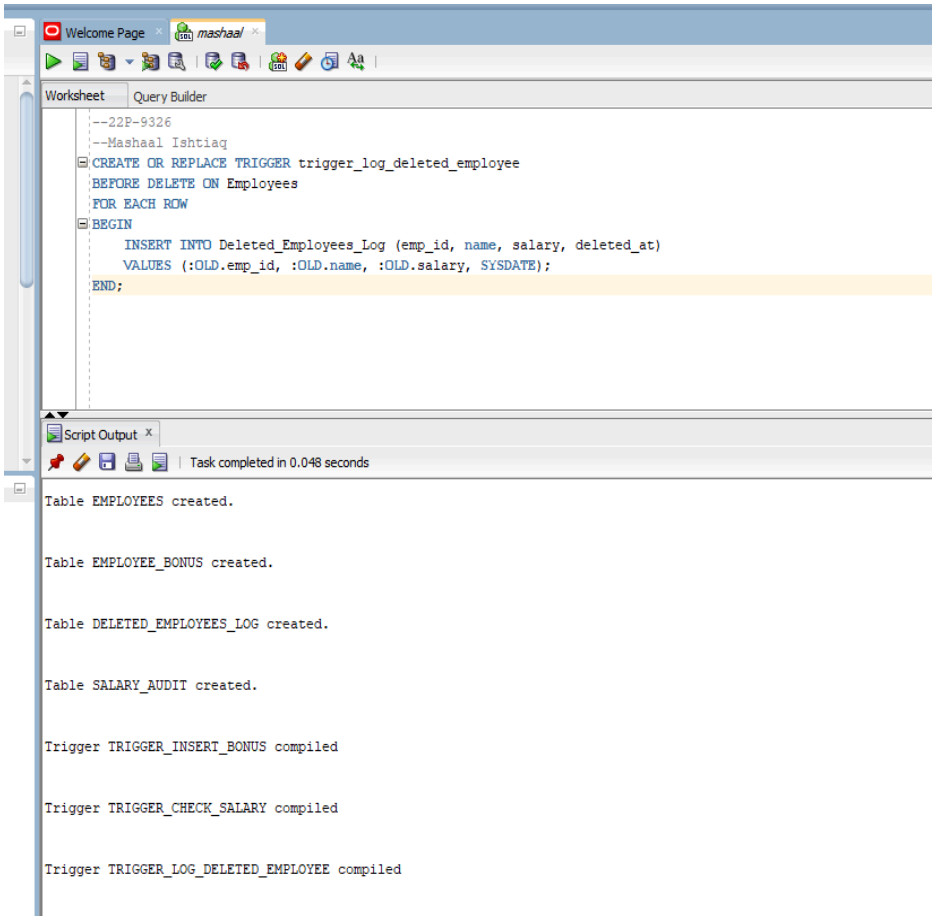

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_insert_bonus
AFTER INSERT ON Employees
FOR EACH ROW
BEGIN
INSERT INTO Employee_Bonus(emp_id, bonus)
VALUES(:NEW.emp_id, :NEW.salary * 0.10);
END;
Script Output
Task completed in 0.092 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.
Table DELETED_EMPLOYEES_LOG created.
Table SALARY_AUDIT created.
Trigger TRIGGER_INSERT_BONUS compiled

#2

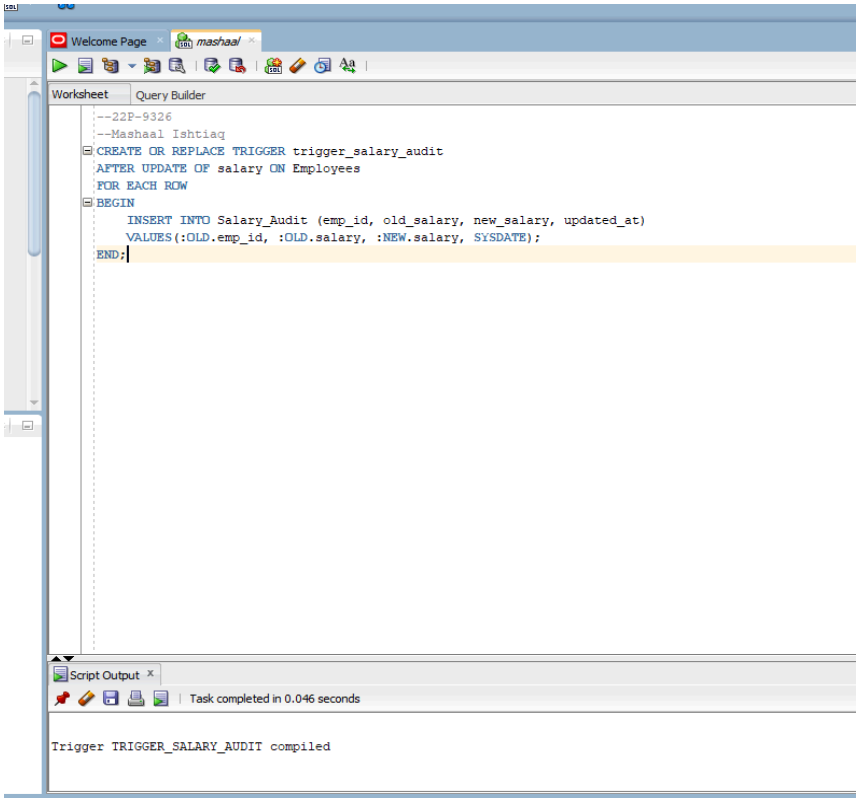

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_check_salary
BEFORE UPDATE OF salary ON Employees
FOR EACH ROW
BEGIN
IF :NEW.salary > 10000 THEN
RAISE_APPLICATION_ERROR(-20001, 'Salary has exceeded the allowed threshold of 10,000.');
END IF;
END;
Script Output
Task completed in 0.038 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.
Table DELETED_EMPLOYEES_LOG created.
Table SALARY_AUDIT created.
Trigger TRIGGER_INSERT_BONUS compiled
Trigger TRIGGER_CHECK_SALARY compiled

#3

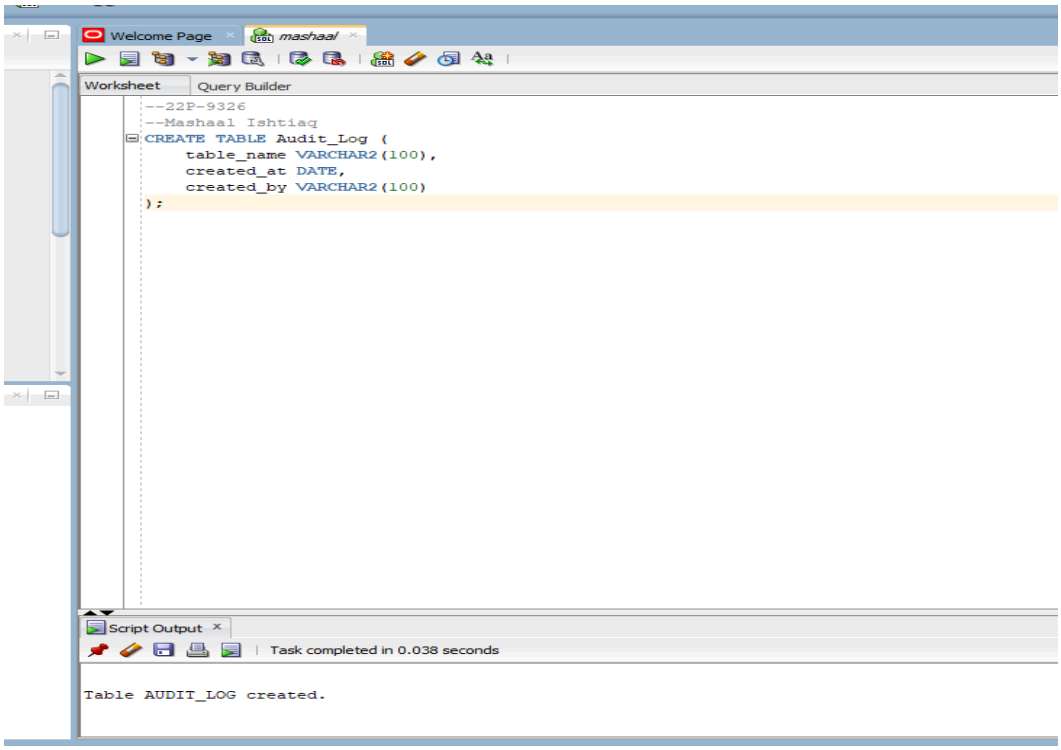

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Deleted_Employees_Log (
emp_id NUMBER,
name VARCHAR2(100),
salary NUMBER,
deleted_at DATE
);
Script Output
Task completed in 0.038 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.
Table DELETED_EMPLOYEES_LOG created.

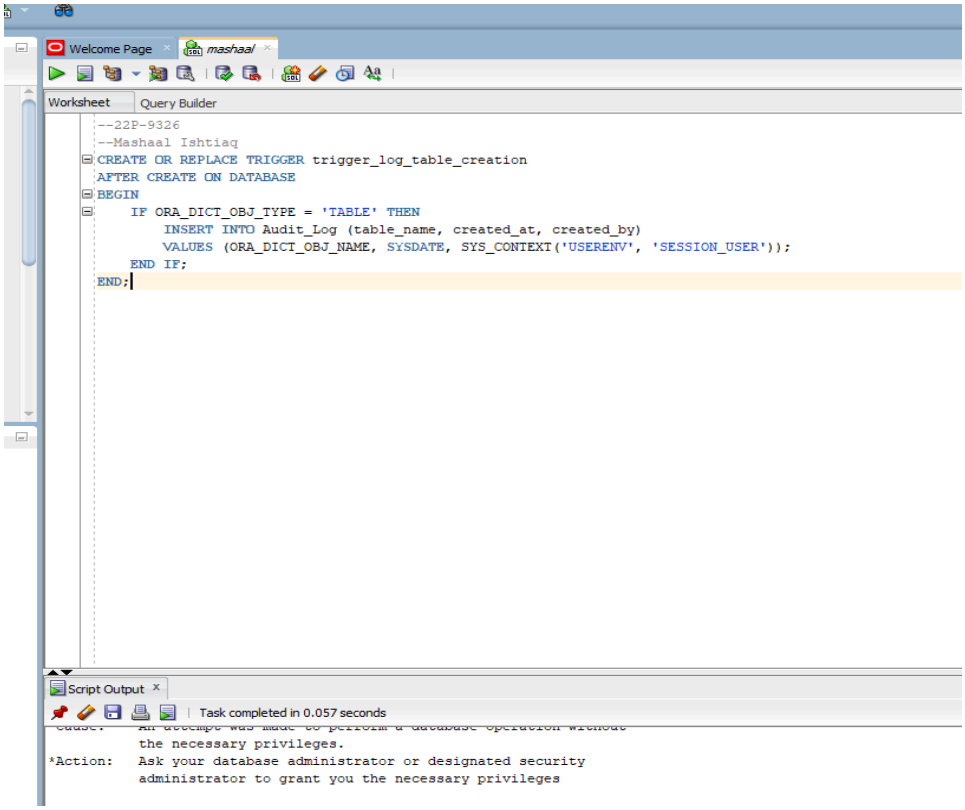

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_log_deleted_employee
BEFORE DELETE ON Employees
FOR EACH ROW
BEGIN
INSERT INTO Deleted_Employees_Log (emp_id, name, salary, deleted_at)
VALUES (:OLD.emp_id, :OLD.name, :OLD.salary, SYSDATE);
END;
Script Output
Task completed in 0.048 seconds
Table EMPLOYEES created.
Table EMPLOYEE_BONUS created.
Table DELETED_EMPLOYEES_LOG created.
Table SALARY_AUDIT created.
Trigger TRIGGER_INSERT_BONUS compiled
Trigger TRIGGER_CHECK_SALARY compiled
Trigger TRIGGER_LOG_DELETED_EMPLOYEE compiled

#4

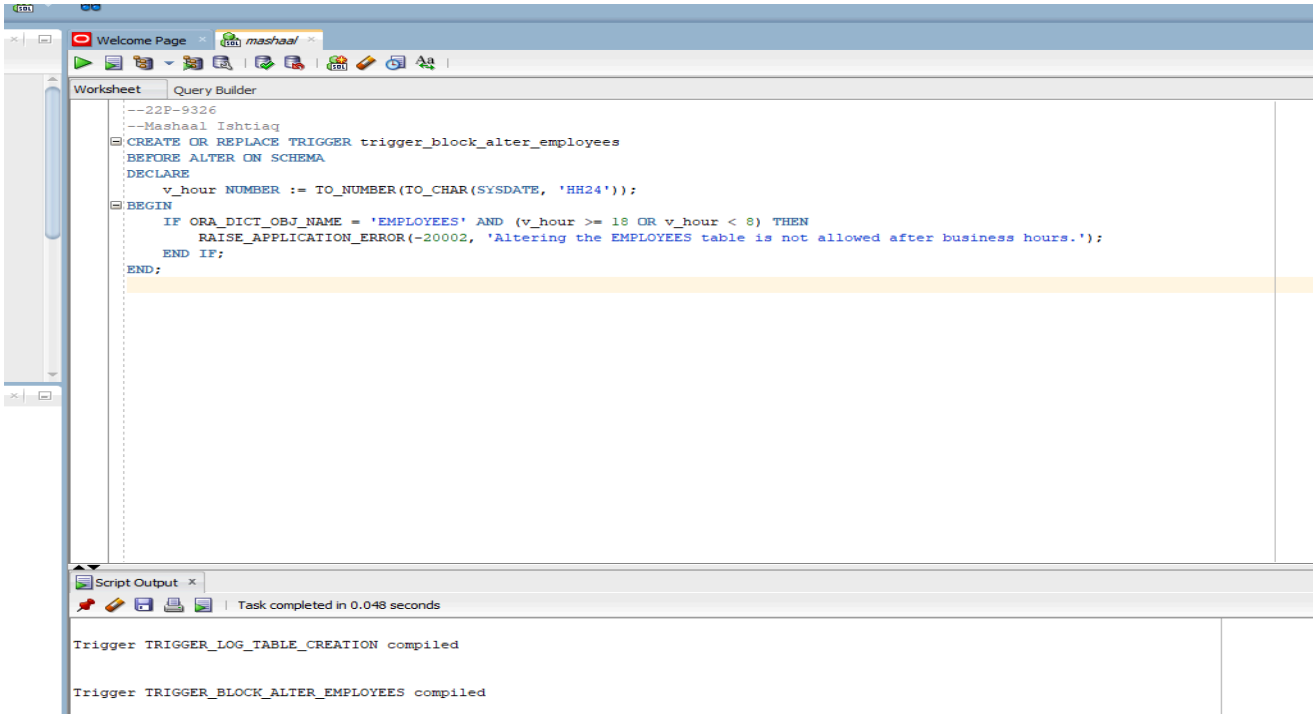

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_salary_audit
AFTER UPDATE OF salary ON Employees
FOR EACH ROW
BEGIN
INSERT INTO Salary_Audit (emp_id, old_salary, new_salary, updated_at)
VALUES(:OLD.emp_id, :OLD.salary, :NEW.salary, SYSDATE);
END;
Script Output
Task completed in 0.046 seconds
Trigger TRIGGER_SALARY_AUDIT compiled

#5

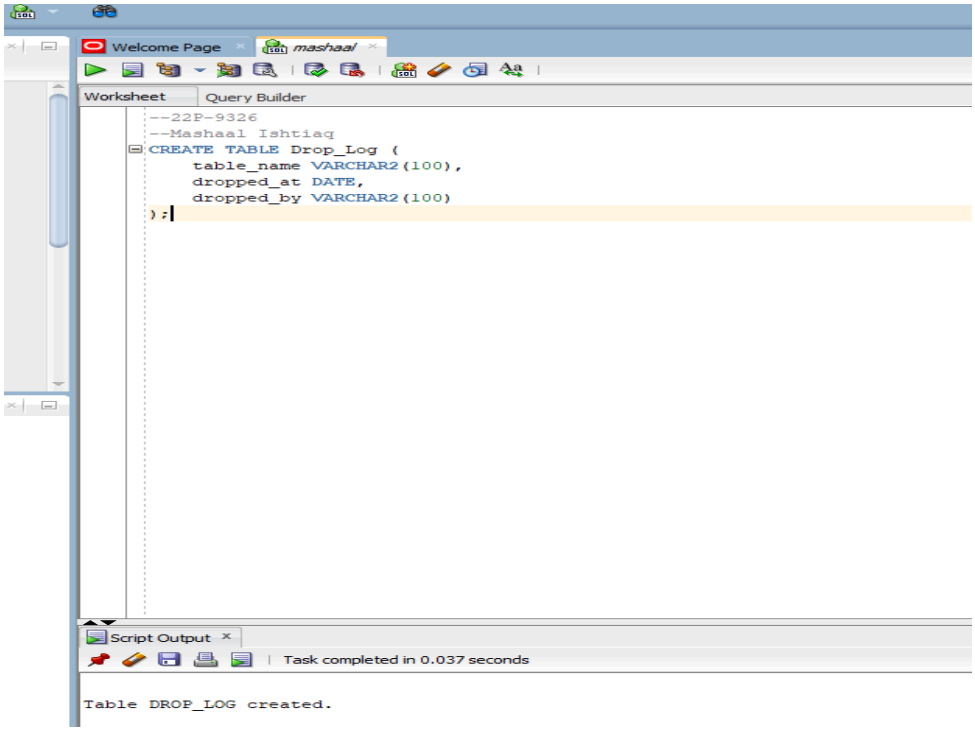

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Audit_Log (
table_name VARCHAR2(100),
created_at DATE,
created_by VARCHAR2(100)
);
Script Output
Task completed in 0.038 seconds
Table AUDIT_LOG created.


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_log_table_creation
AFTER CREATE ON DATABASE
BEGIN
IF ORA_DICT_OBJ_TYPE = 'TABLE' THEN
INSERT INTO Audit_Log (table_name, created_at, created_by)
VALUES (ORA_DICT_OBJ_NAME, SYSDATE, SYS_CONTEXT('USERENV', 'SESSION_USER'));
END IF;
END;
Script Output
Task completed in 0.057 seconds
the necessary privileges.
*Action: Ask your database administrator or designated security
administrator to grant you the necessary privileges

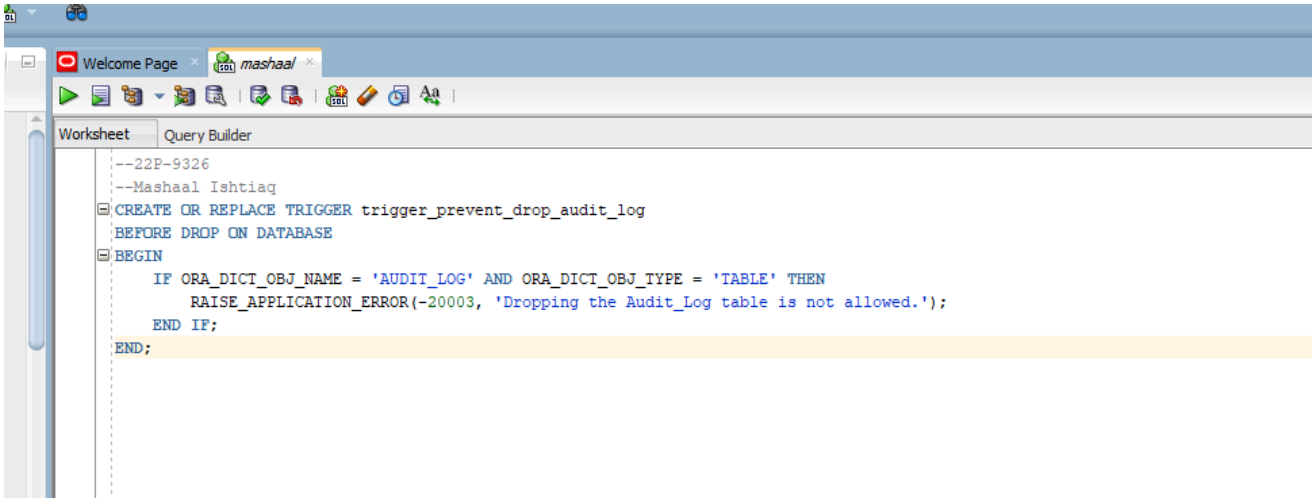
#6


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_block_alter_employees
BEFORE ALTER ON SCHEMA
DECLARE
v_hour NUMBER := TO_NUMBER(TO_CHAR(SYSDATE, 'HH24'));
BEGIN
IF ORA_DICT_OBJ_NAME = 'EMPLOYEES' AND (v_hour >= 18 OR v_hour < 8) THEN
RAISE_APPLICATION_ERROR(-20002, 'Altering the EMPLOYEES table is not allowed after business hours.');
END IF;
END;
Script Output
Task completed in 0.048 seconds
Trigger TRIGGER_LOG_TABLE_CREATION compiled
Trigger TRIGGER_BLOCK_ALTER_EMPLOYEES compiled

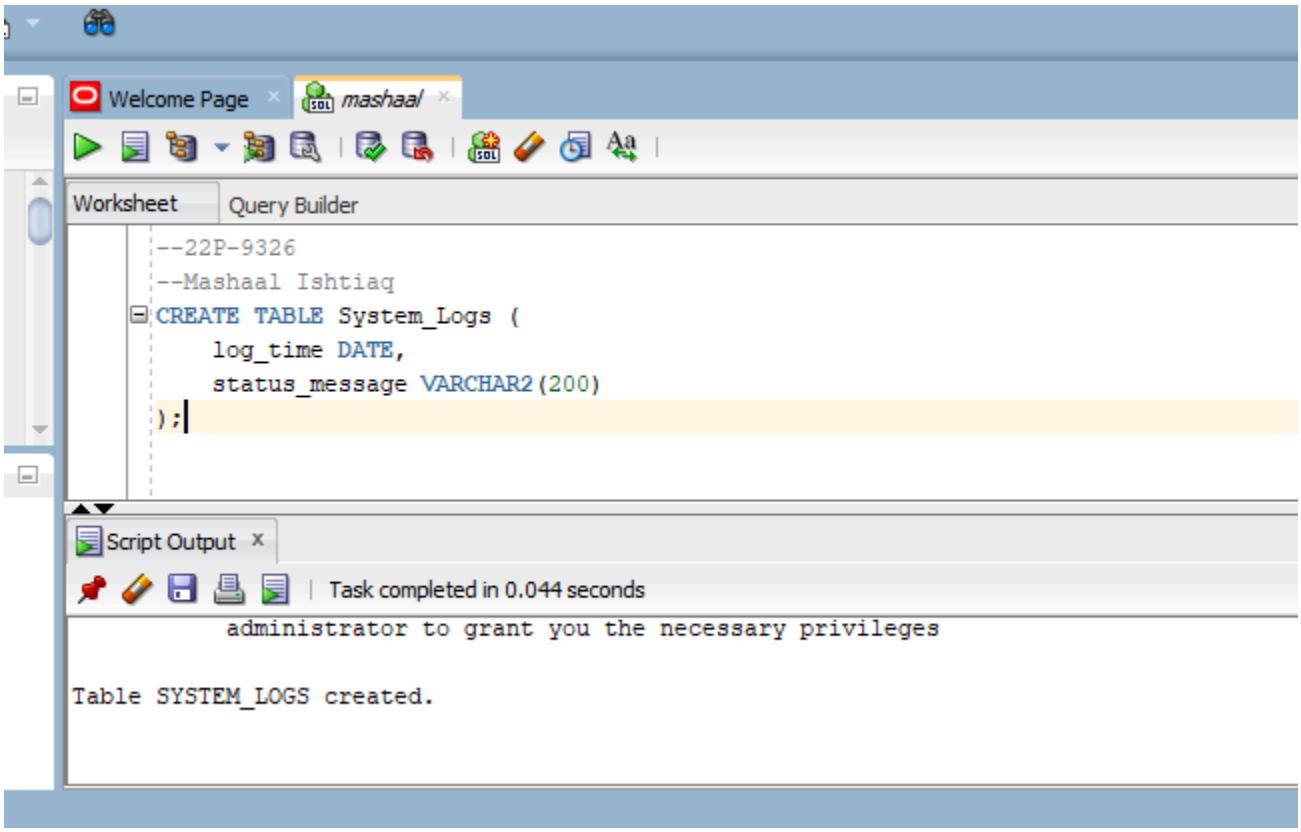
#7

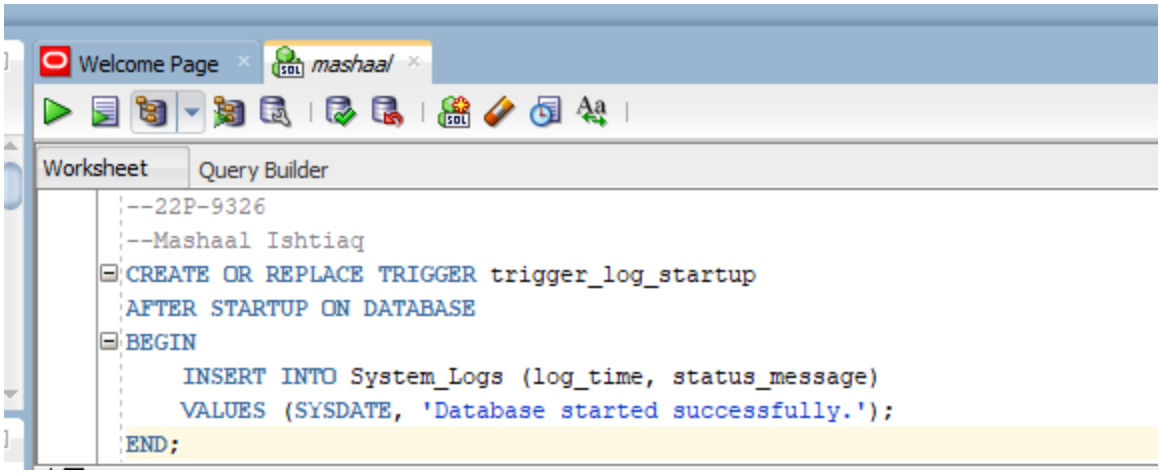

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Drop_Log (
table_name VARCHAR2(100),
dropped_at DATE,
dropped_by VARCHAR2(100)
);
Script Output
Task completed in 0.037 seconds
Table DROP_LOG created.

#8

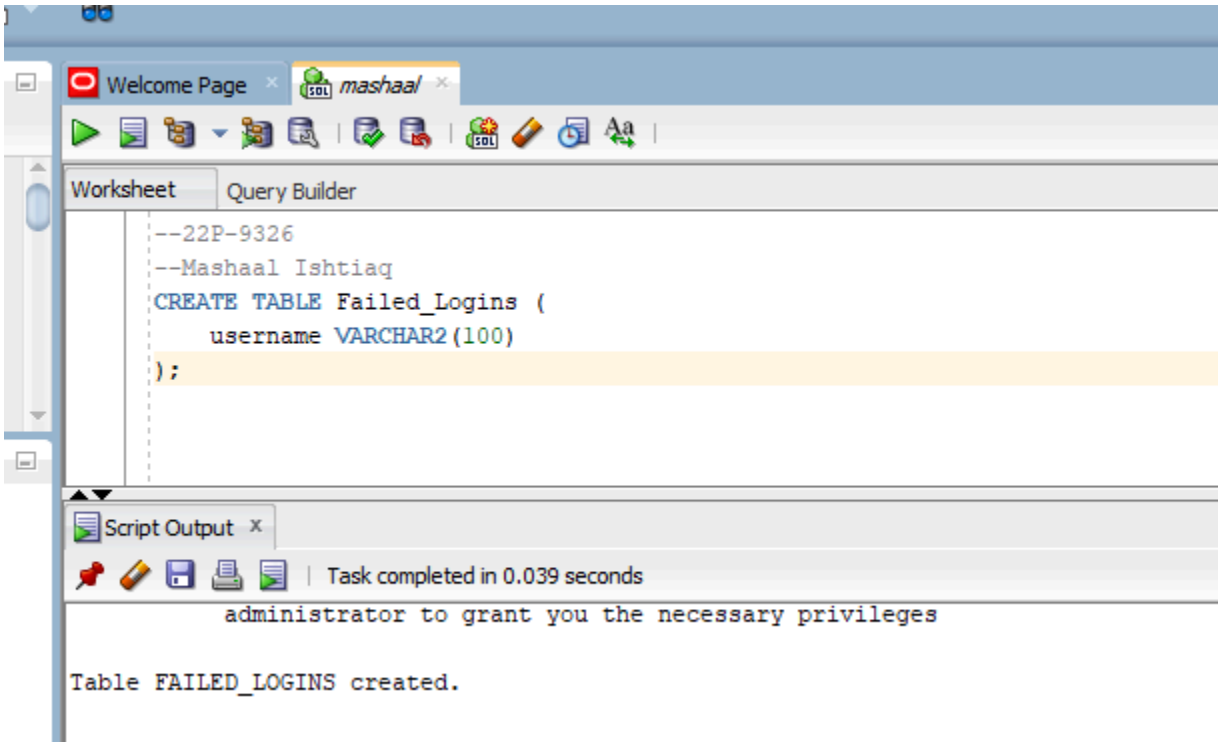

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_prevent_drop_audit_log
BEFORE DROP ON DATABASE
BEGIN
IF ORA_DICT_OBJ_NAME = 'AUDIT_LOG' AND ORA_DICT_OBJ_TYPE = 'TABLE' THEN
RAISE_APPLICATION_ERROR(-20003, 'Dropping the Audit_Log table is not allowed.');
END IF;
END;

#9

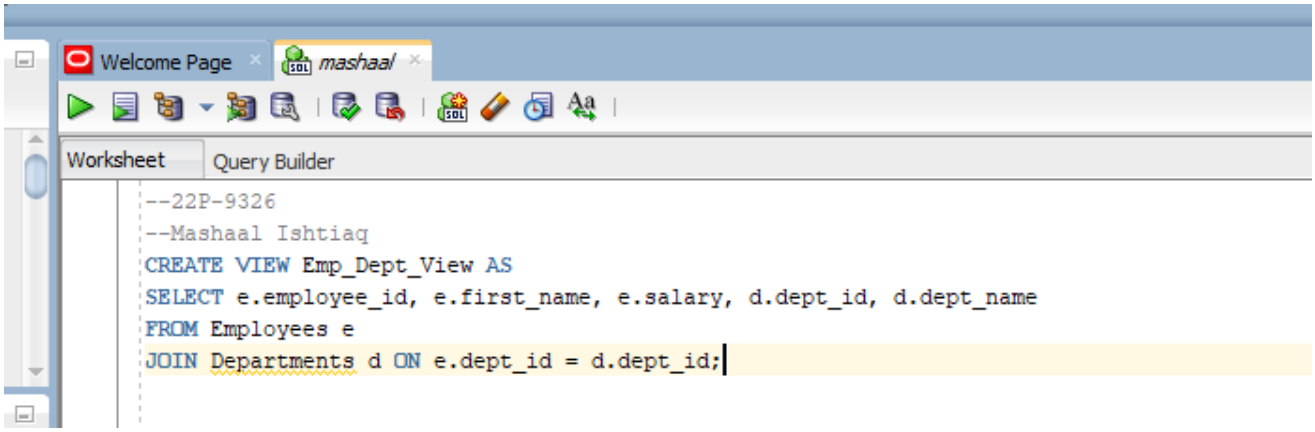

Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE System_Logs (
log_time DATE,
status_message VARCHAR2(200)
);
Script Output
Task completed in 0.044 seconds
administrator to grant you the necessary privileges
Table SYSTEM_LOGS created.


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE OR REPLACE TRIGGER trigger_log_startup
AFTER STARTUP ON DATABASE
BEGIN
INSERT INTO System_Logs (log_time, status_message)
VALUES (SYSDATE, 'Database started successfully.');
END;

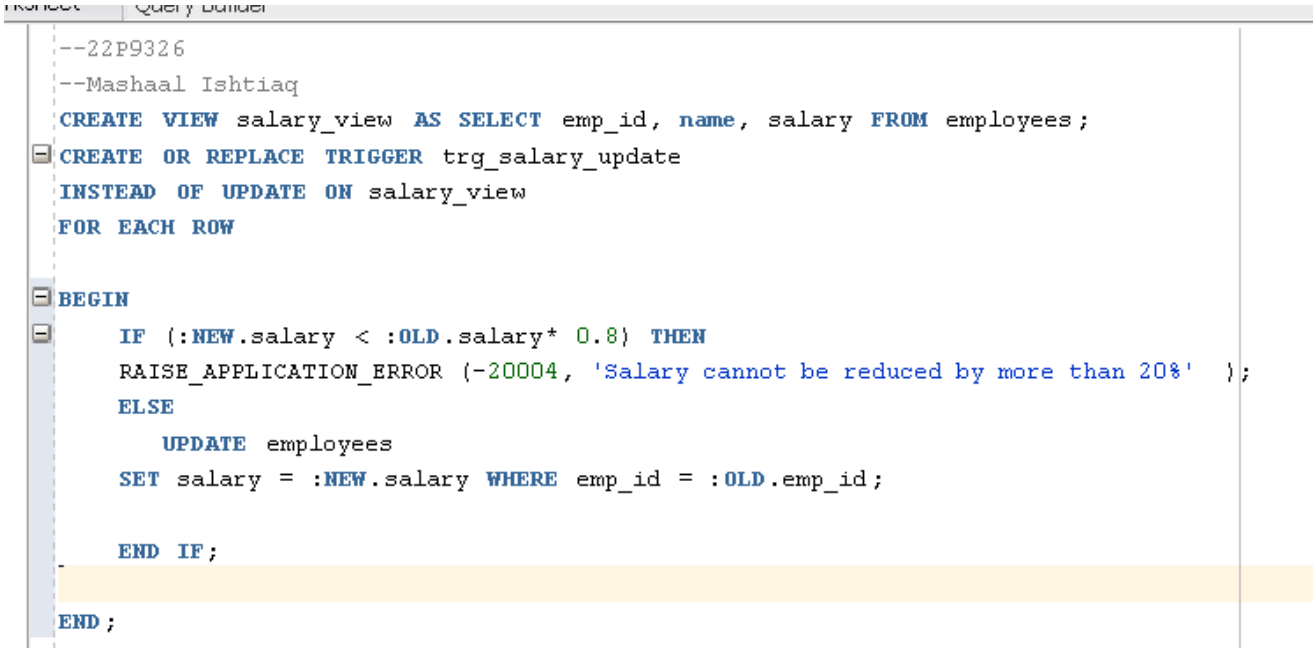
#10


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE TABLE Failed_Logins (
username VARCHAR2(100)
);
Script Output
Task completed in 0.039 seconds
administrator to grant you the necessary privileges
Table FAILED_LOGINS created.

#12


Welcome Page
mashaal
Worksheet
Query Builder
--22P-9326
--Mashaal Ishtiaq
CREATE VIEW Emp_Dept_View AS
SELECT e.employee_id, e.first_name, e.salary, d.dept_id, d.dept_name
FROM Employees e
JOIN Departments d ON e.dept_id = d.dept_id;

#13


--22P9326
--Mashaal Ishtiaq
CREATE VIEW salary_view AS SELECT emp_id, name, salary FROM employees;
CREATE OR REPLACE TRIGGER trg_salary_update
INSTEAD OF UPDATE ON salary_view
FOR EACH ROW
BEGIN
IF (:NEW.salary < :OLD.salary* 0.8) THEN
RAISE_APPLICATION_ERROR (-20004, 'Salary cannot be reduced by more than 20%');
ELSE
UPDATE employees
SET salary = :NEW.salary WHERE emp_id = :OLD.emp_id;
END IF;
END;